

Fixed-yield injection and recovery

2021 10%: independent pilot, native MC and Gaussian signals

Emrys Peets — 24 September 2026

Result and scope

The study completed all 80 evaluation cells with 100 attempted toys per cell: 8,000/8,000 valid signed free fits and 8,000/8,000 valid truth-fixed profiles. The independent pilot contains 2,000/2,000 valid fits. There are 0 unresolved fit/profile rows in the failure ledger. The same expected full-selected signal yield $A = zs_0(m)$ is used for both shapes and every evaluation toy at each mass and level. The Gaussian pilot mean returned yield error defines the frozen s_0 ; z is an input reference-strength label, not an achieved significance. At $z = 5$, mean raw recovery ranges over the ten masses are 0.94–1.10 for Gaussian signals and 0.78–1.04 for native MC. Mean paired response, which subtracts each toy’s matched null fit, ranges from 0.96–0.98 and 0.72–0.89, respectively. These ranges describe cells, not independent pooled measurements. The native-MC paired response is lower than the Gaussian response at every mass at $z = 5$; the pointwise paired-difference intervals below quantify the difference. Equal expected candidate counts do not imply equal extraction difficulty.

m [MeV]	Raw recovery		Paired response		Pull width	
	Gaussian	MC	Gaussian	MC	Gaussian	MC
60	1.04	0.87	0.96	0.72	0.92	1.04
80	1.10	1.04	0.98	0.87	1.08	0.94
100	1.05	0.93	0.98	0.88	0.95	0.87
120	0.99	0.88	0.97	0.88	0.96	1.11
140	0.94	0.91	0.97	0.89	0.96	1.00
160	0.97	0.87	0.97	0.88	0.79	0.87
180	0.98	0.87	0.97	0.88	1.06	1.02
200	0.95	0.82	0.97	0.87	0.93	0.93
220	1.00	0.91	0.97	0.85	0.99	0.94
240	0.95	0.78	0.97	0.81	0.94	0.97

Summary at $z = 5$. Recovery and response are averages of per-toy ratios. Full cell uncertainties and paired differences are supplied as CSV; figures below show standard errors. Pull width is the sample standard deviation, with $n - 1$ denominator.

This is a conditional study of one pinned background mean, fixed empirical MC templates and one archived-kernel GP prescription. It measures fixed-yield bias, error response and nominal signed profile-set containment. It does not establish physical-background adequacy, unconditional coverage, calibrated discovery significance or an exclusion.

Ensemble and frozen reference

Two disjoint cohorts each contain 100 full-support Poisson background spectra, sampled from the bundled nominal 2021 GP arithmetic mean. Each cohort reuses its backgrounds across all ten masses. Evaluation also reuses each background across both shapes and four levels. The 8,000 evaluation rows are therefore correlated across cells; they do not represent 8,000 independent background experiments.

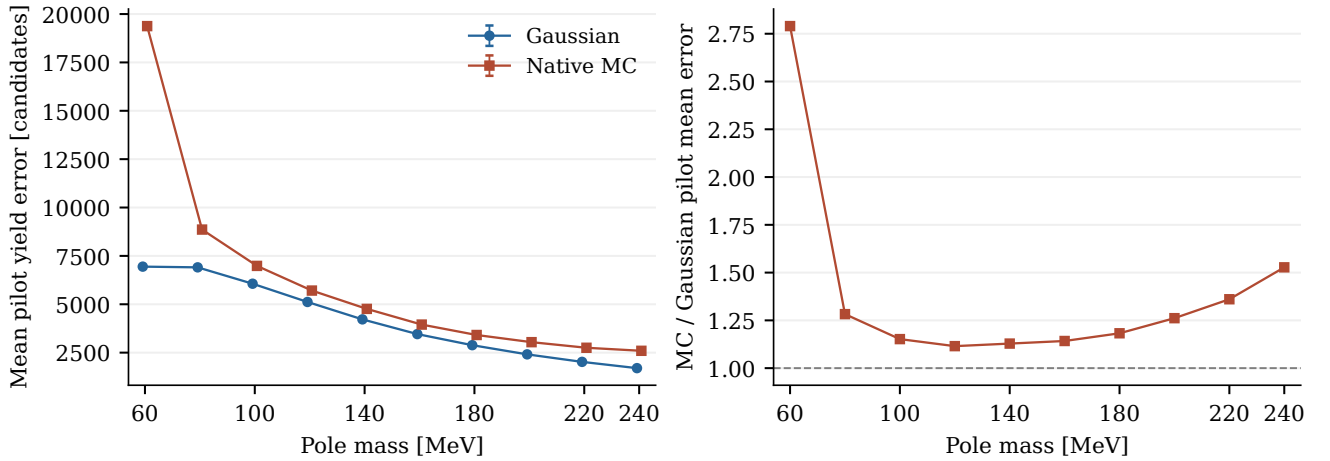
$$B_{ji}^{\text{pilot}}, B_{ti}^{\text{eval}} \sim \text{Poisson}(b_i),$$

$$A(m, z) = z s_0(m),$$

$$s_0(m) = \frac{1}{100} \sum_{j=0}^{99} \hat{\sigma}_{A,G,j}^{\text{pilot}}(m),$$

$$S_{tmzi} \sim \text{Poisson}(A(m, z) w_i).$$

The full probability vector includes bins below and above analysis support. Signal is added in all analysis bins, including GP training sidebands. The realized full signal total fluctuates around A ; expected and realized counts are recorded separately. Within each shape case, generation and extraction use identical full-selected-count probabilities. No fit-window or support renormalization is applied.



Pilot mean returned yield errors (left; bars are standard errors of the 100 pilot errors) and their MC/Gaussian ratio (right). The Gaussian mean alone defines the common frozen yield scale. The ratio panel is descriptive; its sampling uncertainty is not shown.

m [MeV]	s_0	$\text{SE}(s_0)$	Gaussian CV [%]	MC/Gaussian	$A(z = 5)$
60	6944.94	0.38	0.054	2.790	34724.72
80	6907.96	0.16	0.023	1.283	34539.78
100	6062.10	0.12	0.019	1.152	30310.49
120	5118.65	0.11	0.021	1.115	25593.27
140	4219.16	0.10	0.024	1.129	21095.81
160	3458.43	0.10	0.030	1.142	17292.15
180	2885.38	0.10	0.034	1.182	14426.90
200	2410.70	0.10	0.042	1.262	12053.48
220	2022.51	0.12	0.061	1.361	10112.54
240	1698.53	0.14	0.082	1.528	8492.66

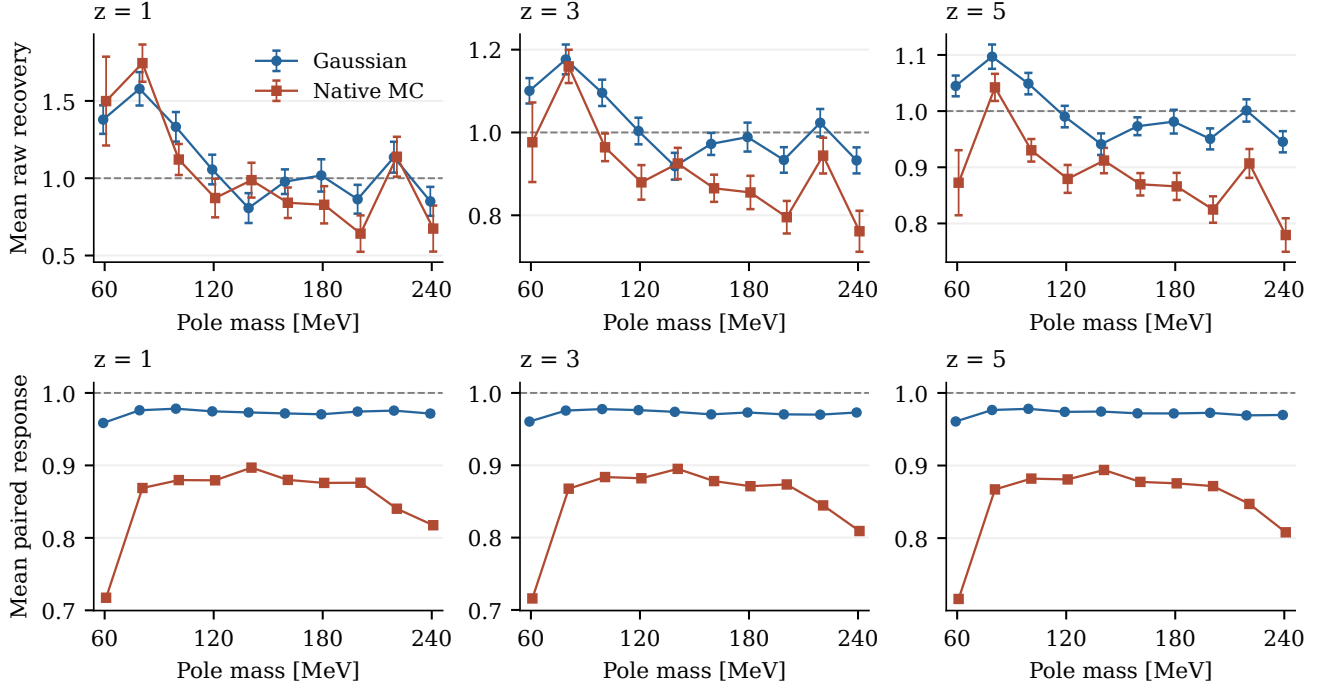
The pilot table is frozen before evaluation. Its standard error describes pilot precision; it is not added to a pull denominator or resampled during evaluation bootstraps.

Raw recovery and paired response

For every positive injection, two diagnostics retain different information:

$$Q_t = \frac{\hat{A}_t(A)}{A}, \quad R_t = \frac{\hat{A}_t(A) - \hat{A}_t(0)}{A}.$$

Raw recovery retains any background-only yield offset. Paired response subtracts the signed null fit of the same background, mass and extraction template. The subtraction is used only here; it is absent from raw yields, pulls and likelihood profiles.



Cell means with standard errors from the toy scatter. The dashed line denotes unit recovery/response. MC and Gaussian receive equal expected full-selected yields at each mass and level; their independent Poisson signal draws need not have equal realized totals.

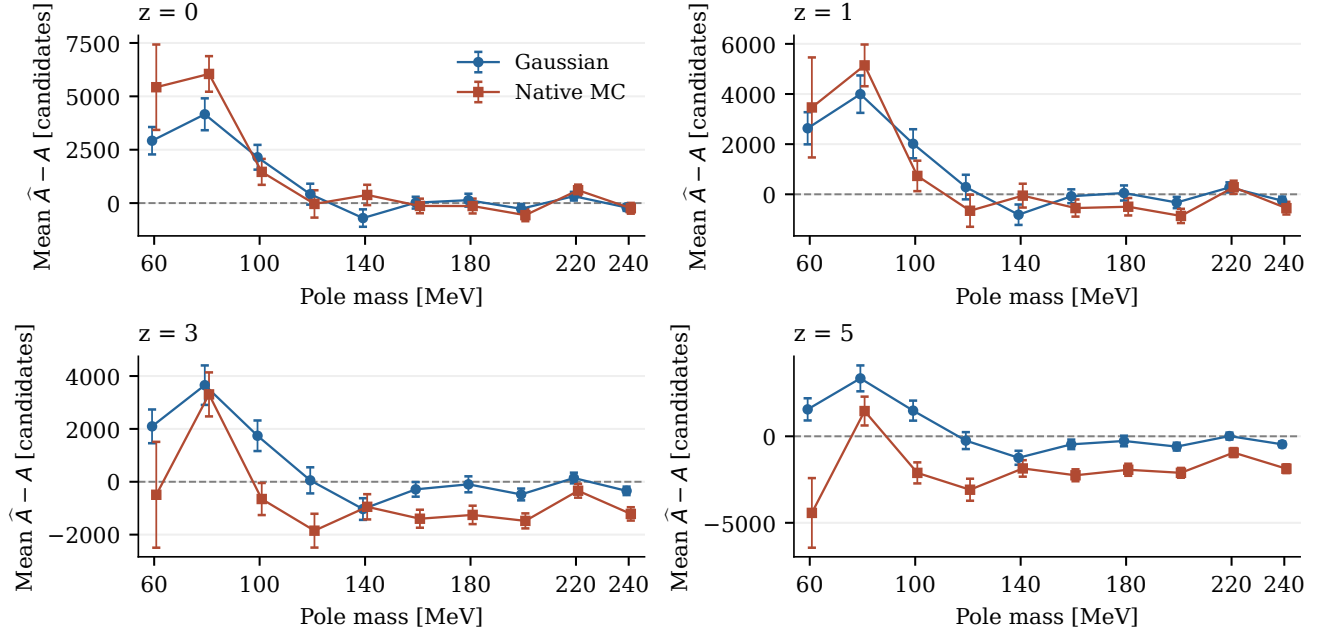
Paired MC–Gaussian response differences at $z = 5$

m [MeV]	MC minus Gaussian	95% bootstrap interval	Complete pairs
60	-0.244	[-0.248, -0.241]	100
80	-0.109	[-0.112, -0.107]	100
100	-0.096	[-0.098, -0.094]	100
120	-0.093	[-0.095, -0.091]	100
140	-0.081	[-0.083, -0.078]	100
160	-0.094	[-0.097, -0.092]	100
180	-0.096	[-0.099, -0.093]	100
200	-0.101	[-0.105, -0.097]	100
220	-0.122	[-0.127, -0.118]	100
240	-0.162	[-0.167, -0.157]	100

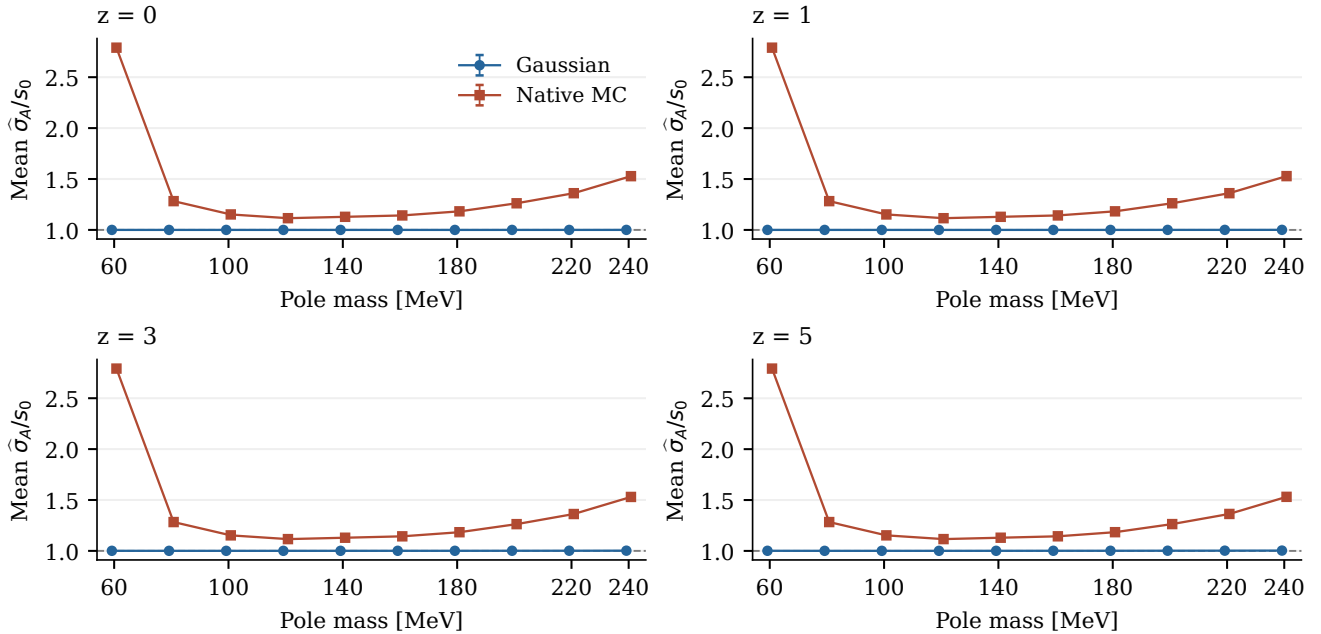
The whole-toy bootstrap uses the same resampled evaluation IDs in every cell and preserves null partners. Each difference uses complete MC/Gaussian pairs; missing pairs are disclosed in the machine-readable table. These are pointwise intervals without a multiple-comparison adjustment.

Absolute bias and returned errors

Absolute bias is $\overline{\hat{A} - A}$ in full-selected candidates. Returned-error response is the average of $\hat{\sigma}_A/s_0$. Neither the returned error nor s_0 is the empirical spread of fitted yields; the latter is saved separately as `Ahat_sd`.



Mean absolute bias; bars are standard errors across valid fits. At $z = 0$ these are the signed null offsets.



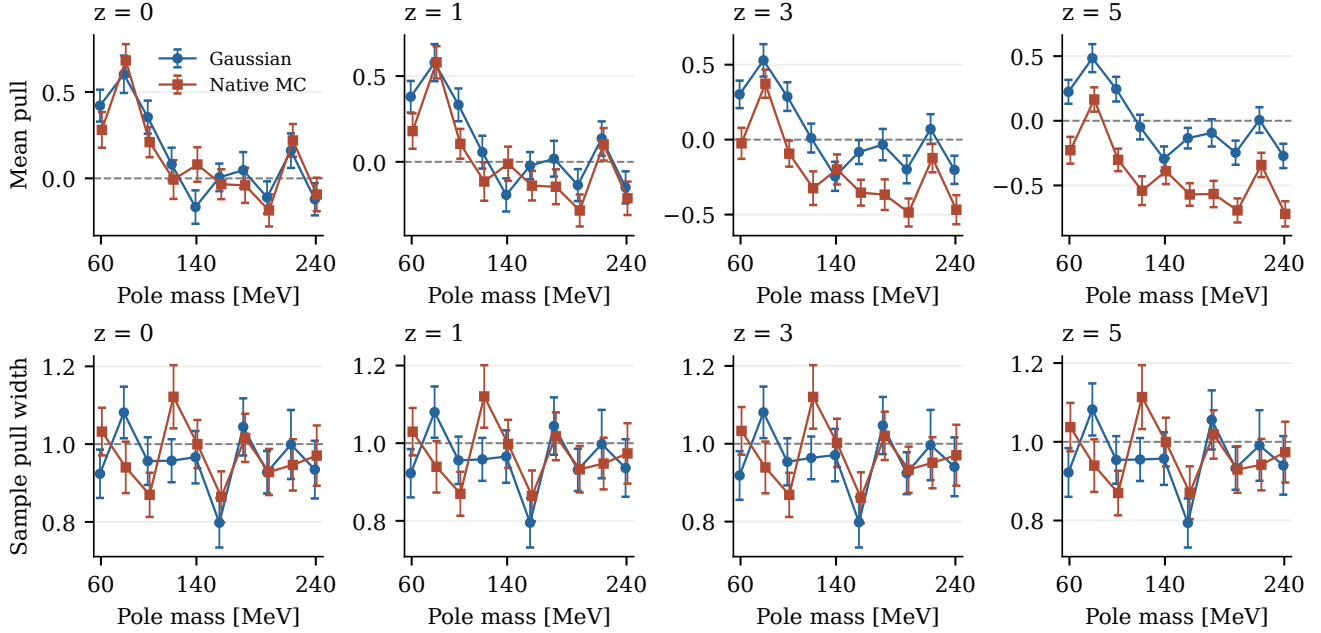
Mean returned yield error divided by the frozen Gaussian reference. Bars are standard errors of the per-toy ratios. The dashed line is unity.

Pull location and width

The primary pull uses the fixed expected yield and each fit's returned profile-Hessian uncertainty:

$$p_t = \frac{\hat{A}_t - A}{\hat{\sigma}_{A,t}}.$$

The realized Poisson signal total is not substituted for A . Unit pull width and zero pull mean are diagnostic reference values, not numerical acceptance criteria.



Top: mean pull with toy-sample standard error. Bottom: sample pull width with one-standard-error whole-toy bootstrap bars (2,000 fixed-seed replicates). Dashed lines mark zero mean and unit width. The accompanying CSV includes pointwise 95% percentile intervals for the widths.

Null offsets cannot be removed by changing the injection scale

At zero signal, each template may project the background residual differently. The null fits remain signed. Using a common expected yield changes the positive-injection ensemble; it does not alter either template's null bias.

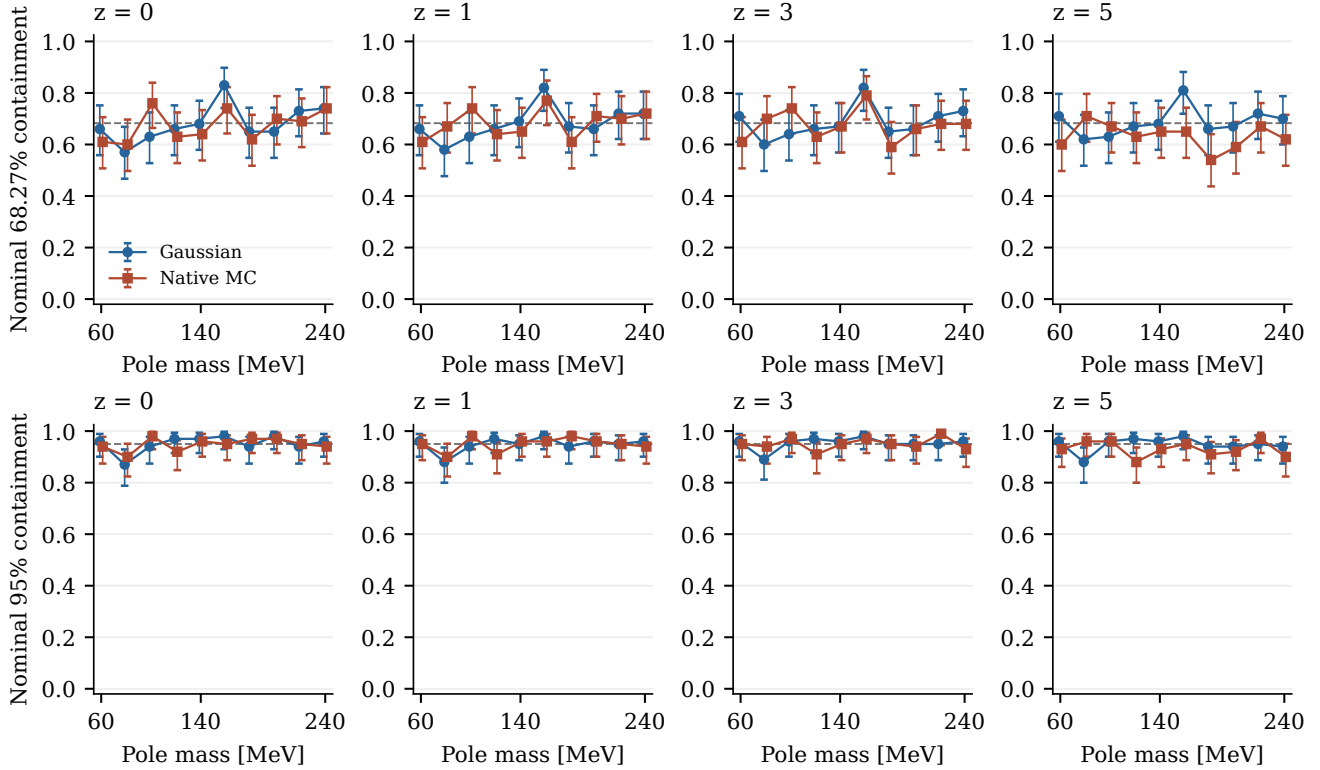
m [MeV]	Null mean pull		Null pull width	
	Gaussian	MC	Gaussian	MC
60	0.42	0.28	0.92	1.03
80	0.60	0.68	1.08	0.94
100	0.35	0.21	0.96	0.87
120	0.08	-0.01	0.96	1.12
140	-0.17	0.08	0.97	1.00
160	0.01	-0.03	0.80	0.86
180	0.05	-0.04	1.04	1.02
200	-0.11	-0.19	0.93	0.93
220	0.16	0.22	1.00	0.95
240	-0.12	-0.09	0.93	0.97

Nominal signed profile-set containment

For each toy, nuisance coordinates are refitted at the fixed expected A , with the same GP mean/covariance used by that toy's free fit:

$$q_{\text{true}} = 2\{\text{NLL}(A, \hat{\theta}_A) - \text{NLL}(\hat{A}, \hat{\theta})\}.$$

Truth is counted inside the nominal 68.27% and 95% signed profile-likelihood sets when $q_{\text{true}} \leq 1$ and $q_{\text{true}} \leq 3.841459$, respectively. These are signed sets, not a physical nonnegative-signal upper-limit construction.



Containment among numerically valid truth profiles. Bars are exact 95% Clopper-Pearson intervals using that cell's valid-profile denominator. Dashed lines show nominal reference levels. Shared backgrounds correlate the cells; the intervals are pointwise.

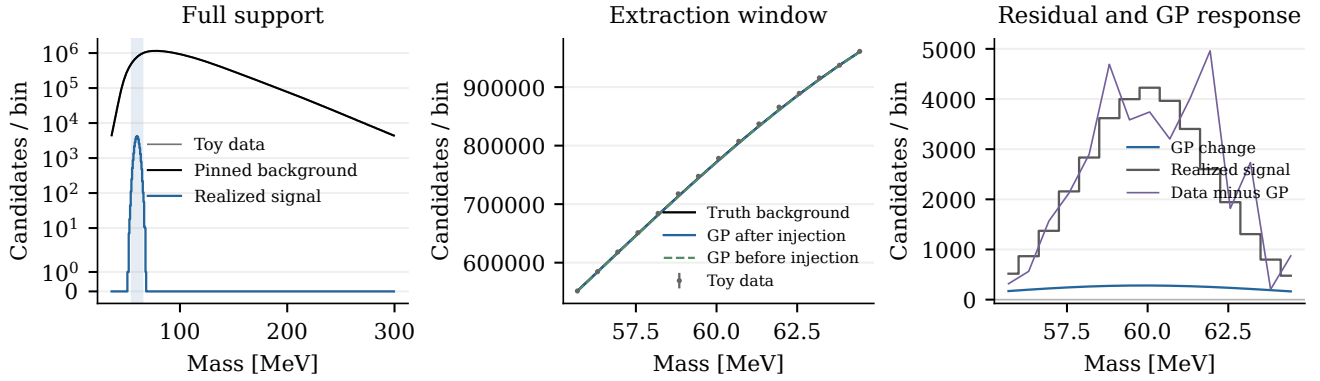
z	attempted/cell	68.27% count range		95% count range	
		Gaussian	MC	Gaussian	MC
0	100	57–83	60–76	87–98	90–98
1	100	58–82	61–77	88–98	90–98
3	100	60–82	59–79	89–98	91–99
5	100	62–81	54–71	88–98	88–97

Ranges are over mass cells; they are not pooled proportions. Exact cell counts, denominators and intervals are in `evaluation_summary.csv`. For k containing profiles and f unresolved decisions, all-attempt accounting bounds are $[k/100, (k + f)/100]$. These bounds are not confidence intervals. Successful-fit containment is conditional on numerical success; the failure ledger and valid fractions remain part of the result. For this run, valid-profile denominators range from 100 to 100 out of 100. With only 100 toys per cell, percentage-level claims require the reported binomial uncertainty; the shared-cell design cannot be treated as additional independent precision.

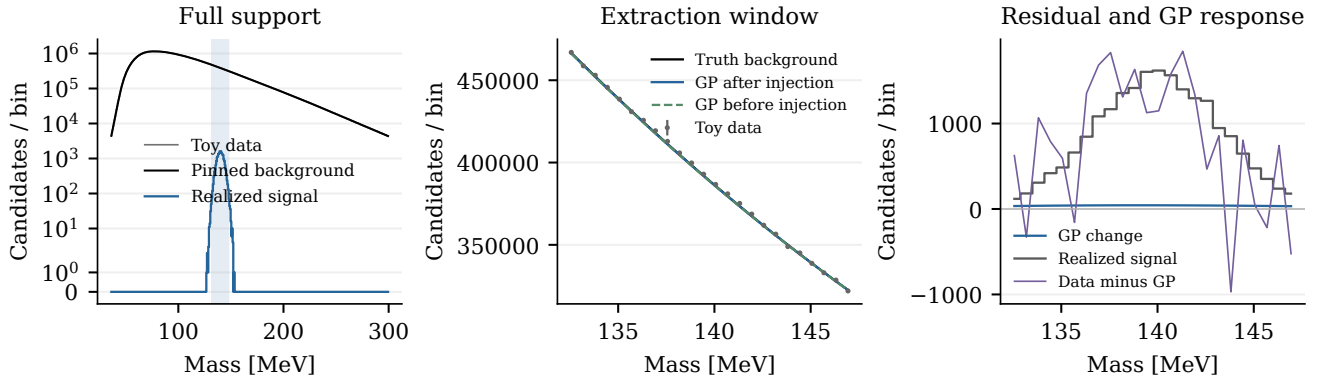
Representative spectra and GP response

Representative spectra are selected by fixed toy ID and declared masses, without selection by fit quality or recovery. Full-support plots show the added signal tails; window and residual panels show the response of GP conditioning to the same injected spectrum. A plotted GP shift is a diagnostic, not a decomposition proving that every recovery difference arises from sideband absorption.

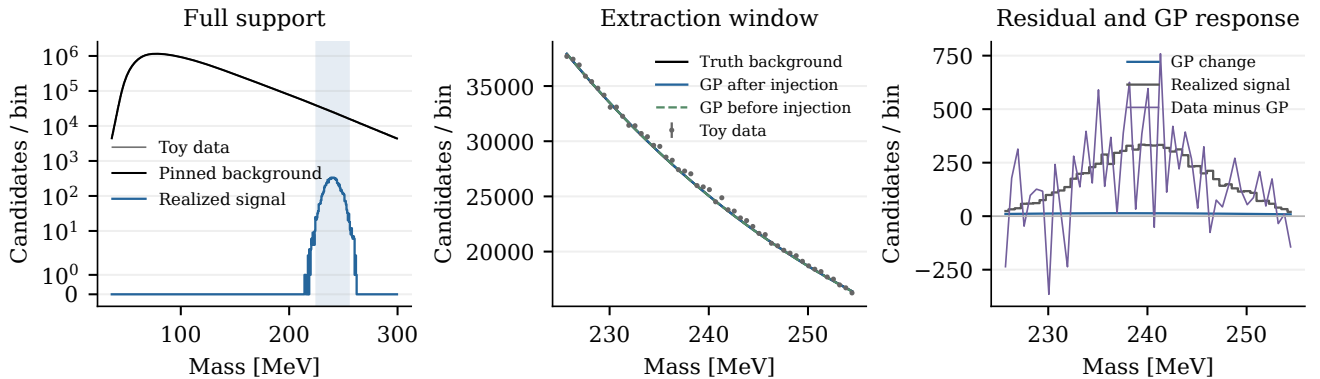
Gaussian signals



60 MeV, Gaussian, $z = 5$, evaluation toy 0. The green dashed curve is the matched background-only GP prediction.



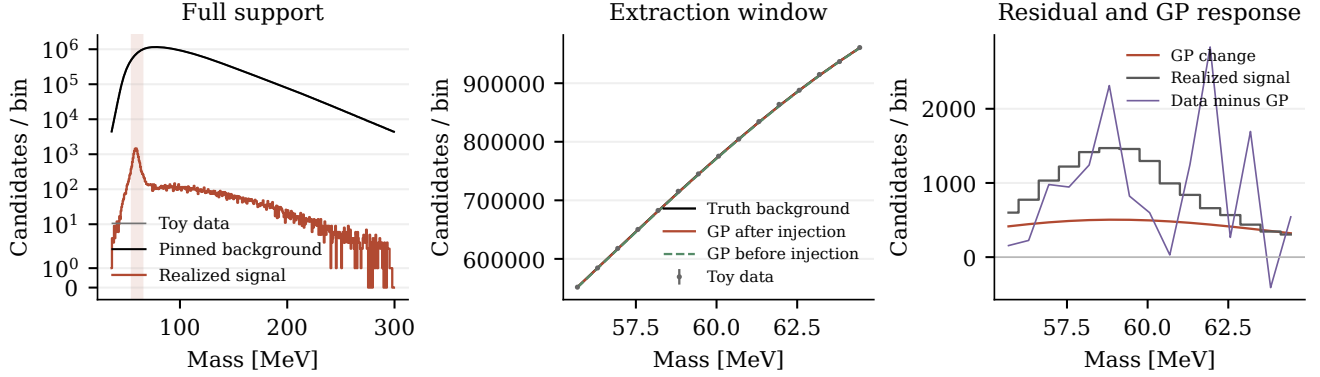
140 MeV, Gaussian, $z = 5$, evaluation toy 0. The green dashed curve is the matched background-only GP prediction.



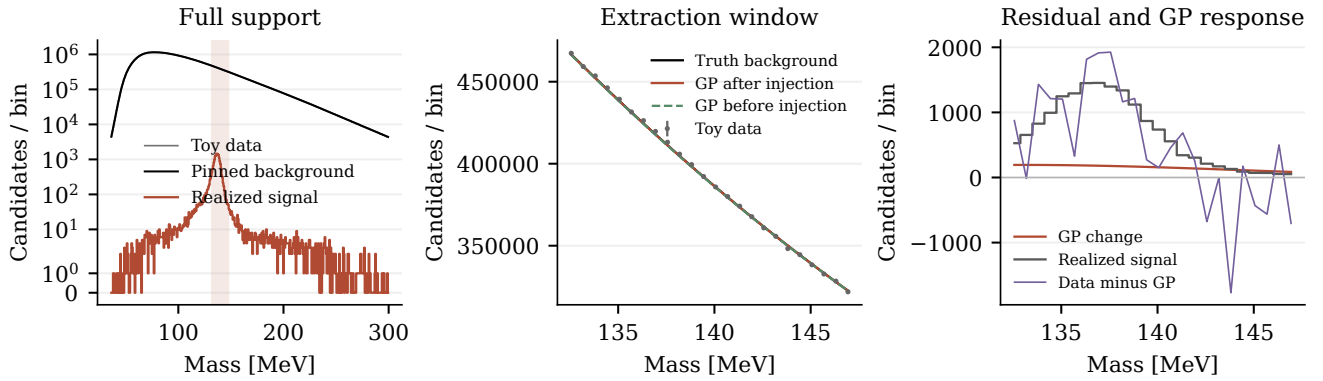
240 MeV, Gaussian, $z = 5$, evaluation toy 0. The green dashed curve is the matched background-only GP prediction.

Representative spectra: native MC

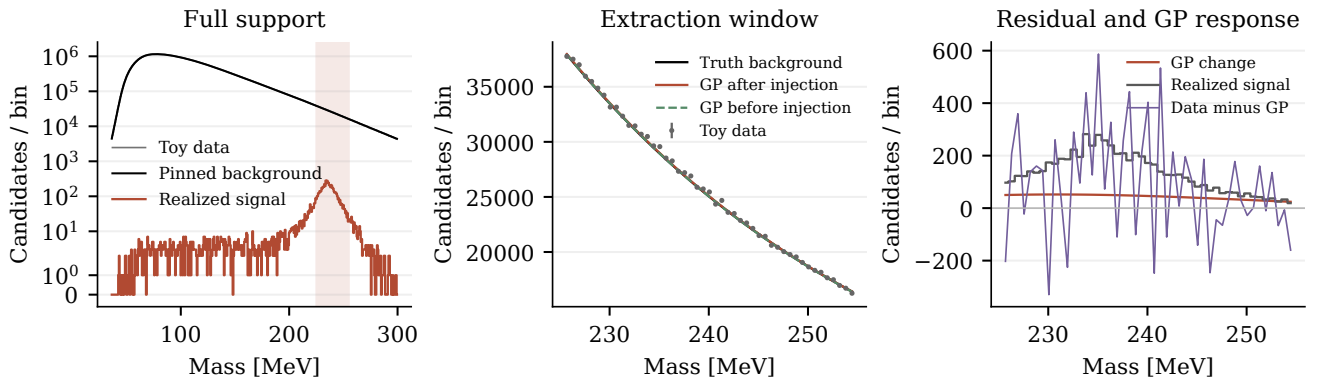
The same evaluation background (toy 0), masses, pole-centered masks and common expected yields are used as on the Gaussian page. Signal draws are independent across shapes. The nominal native-MC template is used both to generate and fit these signals.



60 MeV, Native MC, $z = 5$, evaluation toy 0. The green dashed curve is the matched background-only GP prediction.



140 MeV, Native MC, $z = 5$, evaluation toy 0. The green dashed curve is the matched background-only GP prediction.



240 MeV, Native MC, $z = 5$, evaluation toy 0. The green dashed curve is the matched background-only GP prediction.

Native MC and Gaussian differences can include core offset, width, tails and pole-centered window acceptance. This run does not isolate these components. Clean-sideband or model-mismatch controls would be separately specified diagnostics.

Procedure, numerical accounting and provenance

All masses use the pole-centered window $m \pm 2.25\sigma_m$, with the same extraction and training masks for both shapes. The nominal resolution is $\sigma_m = 0.00184825 - 0.001375m + 0.085875m^2$, with m and σ_m both in GeV. Gaussian bin probabilities are CDF differences; native MC probabilities use the pinned histogram CDF with its inherited uniform-within-bin convention and outside-support categories. The GP uses archived mass-dependent kernel parameters. Each spectrum recomputes its count-dependent log targets/noise, conditional mean and correlated covariance from bins outside the window. Positive counts use $\log n$ and noise variance $1/n$; zero counts use target zero and variance one. This is GP conditioning with fixed kernel parameters, without hyperparameter optimization. The inherited signed count-yield likelihood is

$$\lambda_i = \hat{b}_i + (L\theta)_i + Aw_i, \quad \text{NLL} = \sum_{i \in W} (\lambda_i - n_i \log \lambda_i) + \frac{1}{2} \theta^\top \theta.$$

Poisson means remain positive. Returned uncertainties are observed profile-Hessian yield errors. Truth profiling holds the toy's GP constraint fixed, without further retraining or an extra random GP nuisance draw. Every planned toy ID is retained, including failures. Bounded numerical retries reuse identical saved counts and are selected by convergence and objective value. No toy is accepted by agreement with truth, pull or recovery. Tiny negative likelihood differences are clipped only after the inherited $q_{\text{true}} \geq -2 \times 10^{-6}$ gate. Exact acceptance thresholds, retry starts and watchdog settings are recorded in `protocol.json`; smoke/control checks and numerical attempt records accompany the run.

Randomness and uncertainty. The master seed is 63220260924; namespaces 1, 2, 3 and 4 identify pilot backgrounds, evaluation backgrounds, independent signal draws and bootstrap replicates. Bootstrap uncertainty resamples whole evaluation toy IDs, preserving all masses, shapes, levels and null partners. Means use standard errors from sample scatter. MC-Gaussian contrasts use paired differences and disclose incomplete pairs. The 2,000-replicate bootstrap does not create extra independent toys or change the pilot reference.

Boundary relative to v6.2. This ensemble samples Poisson signal counts at fixed expected yields. The prior v6.2 ensemble sampled exact totals with multinomial bin allocation. They are different experiments; numerical differences cannot be attributed to one code change without matching yields, counts, pairing, templates and fit settings.

Limitations. MC selection equivalence and signal-daughter association are unvalidated. Finite-MC, detector-response and source-estimation uncertainties are not propagated. The generating background is one fixed mean. Containment here does not validate physical-background adequacy or a full hyperparameter-optimized production search, and it supplies no global significance, discovery probability or exclusion.

Reproduction. The package includes the immutable final handoff, input/code hashes, saved pilot/evaluation background cohorts, templates, frozen pilot table, per-toy rows and checkpoints, summary scripts, report source and manifest. See `README.md`, `protocol.json`, `provenance/` and `MANIFEST.sha256`. The final handoff and frozen reference hashes are: Final handoff: 68358b6d4a24b83ddb25617f82064c4a3fc1f5bce4b1f7772a9620a58ff94ad8

Frozen pilot reference: f2d9d4bf804ed99894085ecb907fc28734462d6839d3d32bef52fd866f16fd39

Appendix: offset and response transfer

2016 Gaussian signals at controlled exposure factors 0.1 and 1

Emrys Peets — 24 September 2026

Question, scope and independent cohorts

No automatic exposure transfer or production correction is justified. The simple null-offset scaling law is not exact in this conditional experiment.

Nominal 42 MeV gives $\delta_{100} - 10\delta_{10} = +5977$ candidates (pointwise 95% paired bootstrap interval [3105, 8872]); Stress 76 MeV gives $\delta_{100} - 10\delta_{10} = -7279$ candidates (pointwise 95% paired bootstrap interval [-10911, -3903]).

Same-source calibration can center held-out toys; transfer to another generator requires separate validation. The offset, signal response and finite-grid limits below are conditional on pinned sources, Gaussian generation/extraction and the archived-kernel GP prescription.

The masses are 42, 44, 60, 66, 76, 90, 92, 117, 160 and 178 MeV. Exposure factors $L = 0.1$ and 1 scale the same full-exposure generating mean. They are paired by Poisson thinning within a cohort and source; nominal and stress cohorts are independent. Toy IDs are shared across masses and injection levels. Counts of fits below therefore do not count independent experiments.

Cohort	Sources	z levels	Backgrounds/cell	Free fits
Pilot	Nominal	0	100	2,000
Calibration	Nominal, stress	0, 1, ..., 8	100	36,000
Evaluation	Nominal, stress	0, 1, 3, 5	100	16,000

The saved run contains 2,000/2,000 valid pilot free fits, 36,000/36,000 valid calibration free fits, 16,000/16,000 valid evaluation free fits. Native full-exposure CLs limits are valid for 8,000/8,000 attempts. The failure ledger has 0 rows. Numerical validity is separate from scientific closure.

The nominal pilot defines $s_0(m, L)$, the mean returned Gaussian yield error, independently at each exposure. Signal-bin counts are Poisson with fixed expectation $A = z s_0$ and full-selected Gaussian probabilities. Outside-support loss is retained; training-sideband injection is included; there is no window renormalization. The same expected yield is used for nominal and stress generators. z is a reference-strength label, not a significance.

Why the offset may become more visible

In a local linear approximation, write the mean background residual as Lr and the effective covariance as LV . For a fixed signal vector w ,

$$\delta_L = \frac{w^\top (LV)^{-1} (Lr)}{w^\top (LV)^{-1} w} = L\delta_1, \quad \sigma_L \simeq \sqrt{L}\sigma_1, \quad \frac{\delta_L}{\sigma_L} \simeq \sqrt{L} \frac{\delta_1}{\sigma_1}.$$

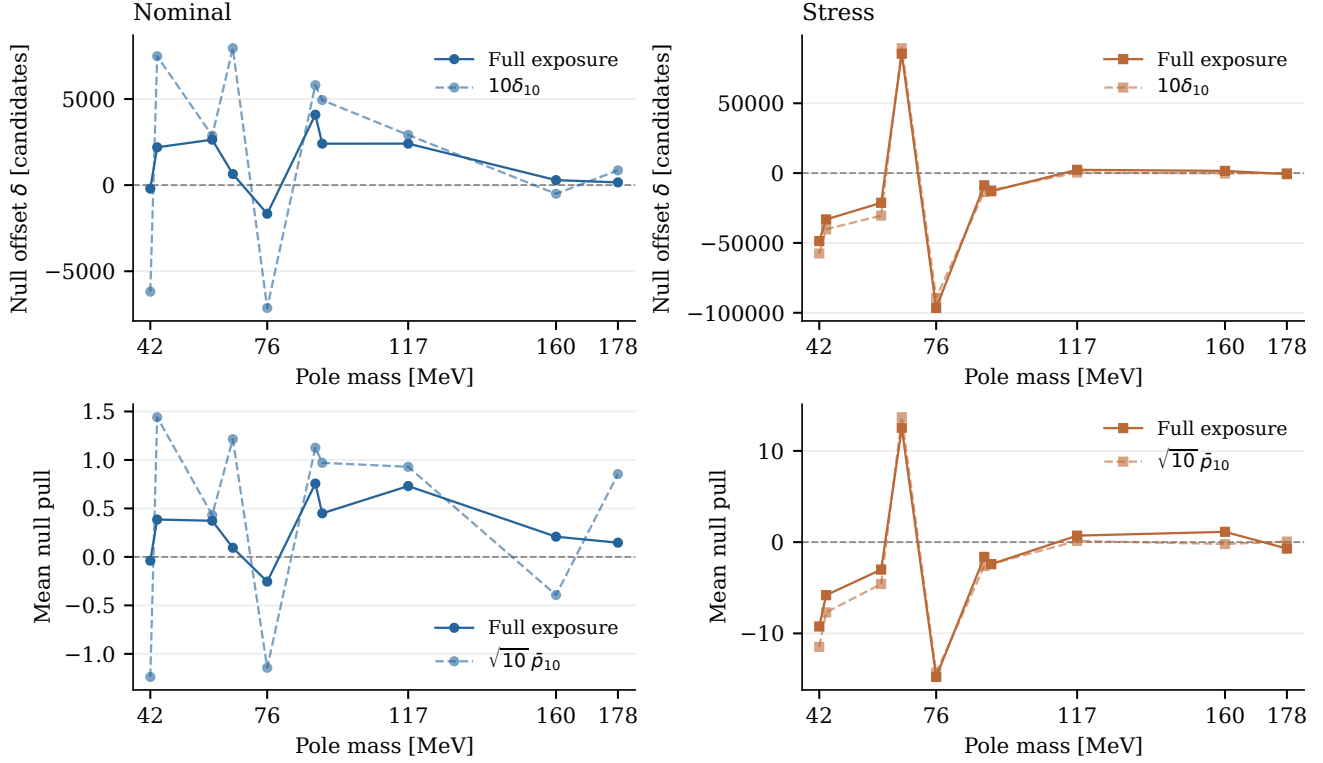
Thus $\delta_{100} \simeq 10\delta_{10}$ and $\bar{p}_{100} \simeq \sqrt{10}\bar{p}_{10}$. These are testable approximations, not identities of the count-dependent GP and nonlinear likelihood. Changing the kernel, mask, generator shape or effective covariance invalidates a simple exposure-only transfer.

The existing nine-page 2021 study is preserved unchanged. This appendix is a separate 2016 Gaussian experiment; it does not revise the earlier MC/Gaussian measurements or turn their diagnostics into calibrated limits.

Practical interpretation. One observed 10% fitted yield or residual is not an ensemble bias: it includes counting noise and possible model mismatch. Transferring it by a factor of ten is not justified by the exposure label alone. A production correction requires qualified background sources, selection-equivalent historical input and independent full-exposure validation, with offset, response and width uncertainties propagated. This appendix changes no production limits.

Exposure scaling of the null offset

Calibration null fits define $\hat{\delta}_L = \widehat{A}_L(0)$. The null pull is $p = \hat{A}/\hat{\sigma}_A$. Because the returned errors fluctuate, the mean pull need not equal $\hat{\delta}_L/\hat{\sigma}_A$. Exposure differences preserve the nested background pairing.



Calibration measurements. Solid: full exposure. Dashed: the 10% estimate transferred using the stated factor. Nominal (blue) and stress (orange) are separate fixed generating means.

m [MeV]	Nominal $\Delta\delta$	Nominal $\Delta\bar{p}$	Stress $\Delta\delta$	Stress $\Delta\bar{p}$
42	5976.8 ± 1466.6	1.20 ± 0.29	8866.6 ± 1452.0	2.24 ± 0.29
44	-5290.3 ± 1549.9	-1.06 ± 0.30	7018.0 ± 1471.6	1.89 ± 0.28
60	-228.2 ± 1945.0	-0.06 ± 0.29	9181.9 ± 1971.9	1.57 ± 0.30
66	-7302.5 ± 2094.0	-1.12 ± 0.32	-3946.8 ± 1983.6	-1.16 ± 0.30
76	5468.3 ± 1692.9	0.89 ± 0.27	-7279.1 ± 1795.9	-0.46 ± 0.29
90	-1716.3 ± 1521.5	-0.37 ± 0.29	4758.1 ± 1461.6	1.00 ± 0.28
92	-2530.9 ± 1494.3	-0.52 ± 0.29	-722.3 ± 1379.9	-0.02 ± 0.27
117	-502.9 ± 818.6	-0.20 ± 0.26	1919.0 ± 917.6	0.58 ± 0.29
160	793.8 ± 401.3	0.60 ± 0.32	1859.9 ± 372.8	1.33 ± 0.29
178	-700.5 ± 346.9	-0.71 ± 0.34	-789.6 ± 310.0	-0.76 ± 0.30

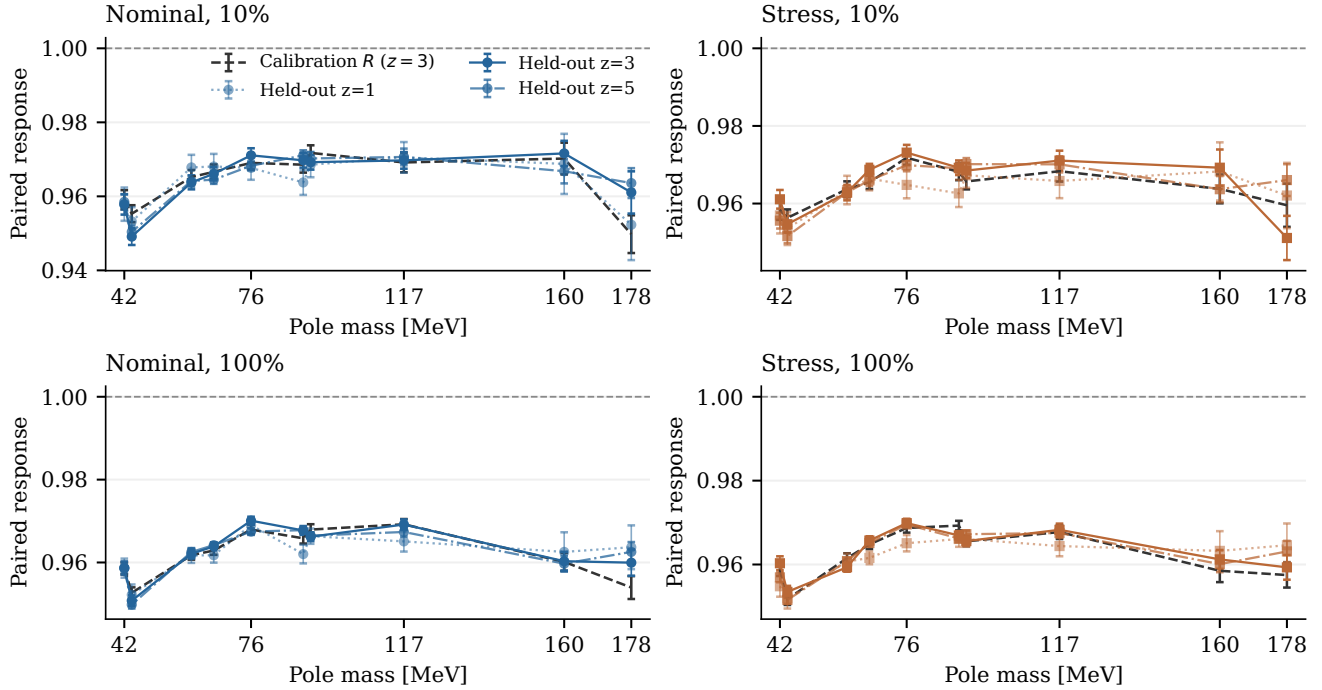
$\Delta\delta = \delta_{100} - 10\delta_{10}$ (candidates), $\Delta\bar{p} = \bar{p}_{100} - \sqrt{10}\bar{p}_{10}$. Errors are pointwise standard errors; the paired bootstrap intervals are retained in `scaling_comparisons.csv`.

Freeze the response before testing the correction

At each source, exposure and mass, the independent calibration cohort fixes

$$\hat{\delta} = \overline{\hat{A}(0)}, \quad \hat{R} = \frac{\overline{\hat{A}(3s_0) - \hat{A}(0)}}{3s_0}, \quad \tilde{A} = \frac{\hat{A} - \hat{\delta}}{\hat{R}}.$$

The $z = 3$ definition is fixed in advance. Held-out $z = 1, 3, 5$ paired responses test its transfer across strength. A constant offset cancels exactly from paired response; subtracting it cannot restore a response slope below one.



Frozen calibration R (black dashed, standard-error bars) and held-out raw paired responses at three positive injection levels. Bars summarize toy variation; the independent calibration table is not refitted using evaluation toys.

Source	Exposure	Frozen R	$z = 1$	$z = 3$	$z = 5$
Nominal	10%	0.950–0.972	0.952–0.971	0.949–0.972	0.950–0.971
Nominal	100%	0.953–0.969	0.952–0.969	0.951–0.970	0.950–0.968
Stress	10%	0.956–0.972	0.954–0.968	0.951–0.973	0.952–0.970
Stress	100%	0.952–0.969	0.952–0.966	0.953–0.970	0.952–0.970

Ranges span the ten correlated mass cells; they are not pooled estimates. Full cell values, complete-pair counts and uncertainty are in the CSV tables.

Source	z	Raw	Transfer offset	Direct offset	Affine
Nominal	0	[-0.48, 0.85]	[-1.06, 1.16]	[-0.22, 0.09]	[-0.22, 0.09]
Nominal	3	[-0.57, 0.75]	[-1.17, 1.04]	[-0.31, -0.01]	[-0.22, 0.10]
Stress	0	[-14.52, 12.52]	[-0.86, 1.59]	[-0.21, 0.25]	[-0.21, 0.25]
Stress	3	[-14.60, 12.41]	[-0.95, 1.47]	[-0.30, 0.16]	[-0.21, 0.25]

Full-exposure mean-pull ranges across mass cells, using each source’s own frozen calibration. Brackets also denote ranges, not confidence intervals.

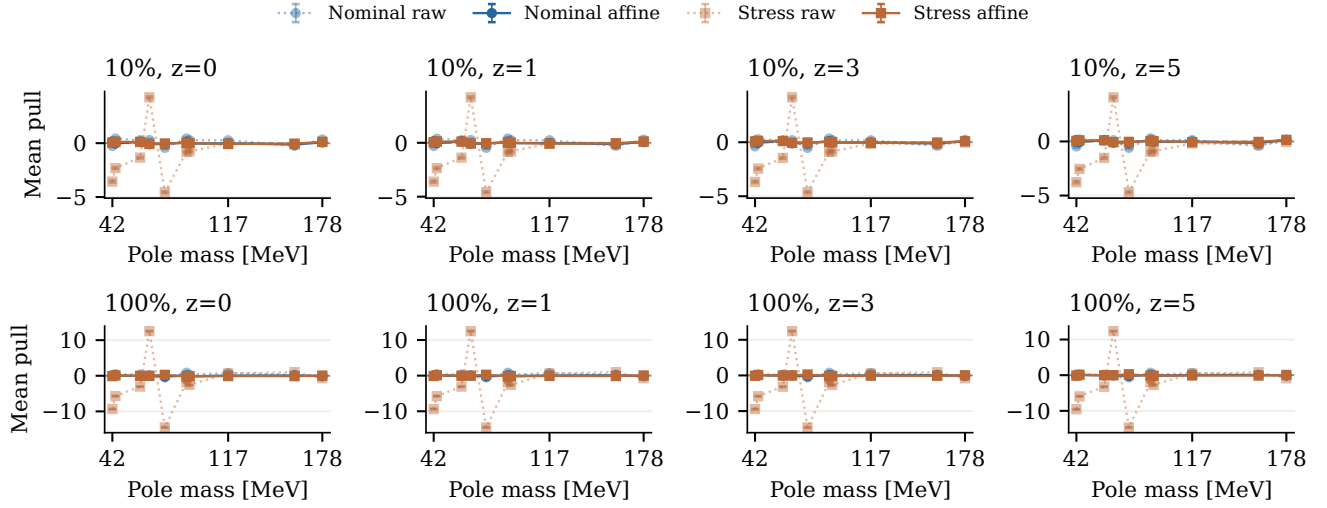
The offset-transfer diagnostic uses $10\hat{\delta}_{10}$ at full exposure; direct-offset subtraction uses the full-exposure calibration offset. The affine diagnostic also divides by the corresponding calibrated response. None of these corrected estimators is inserted into the native profile likelihood or used to rescale a published upper limit.

Held-out pulls after the frozen correction

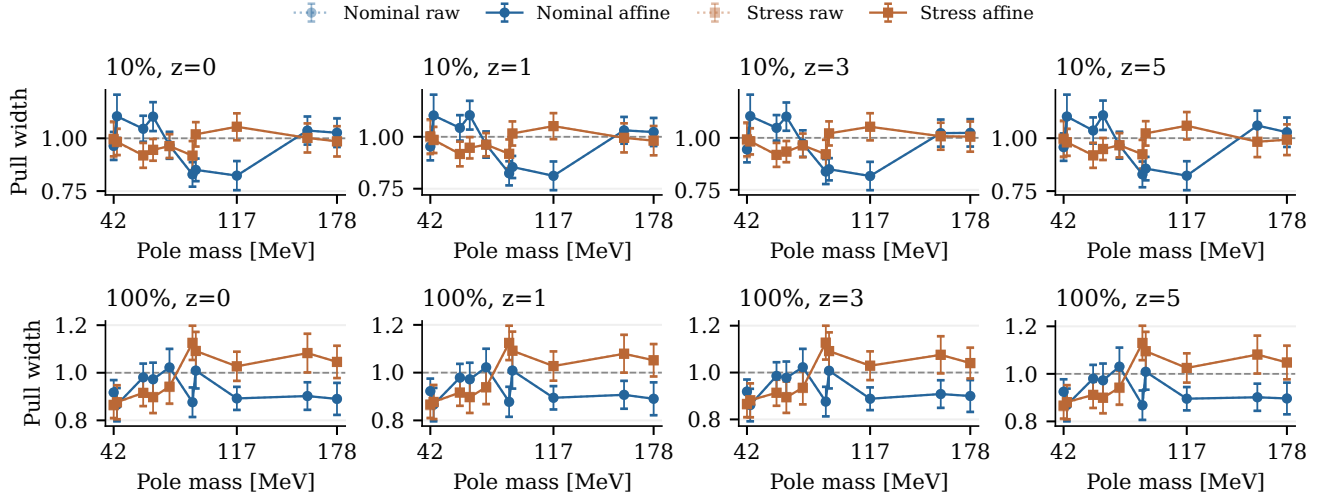
For positive $\hat{R}$, the statistical-only corrected error and pull are

$$\hat{\sigma}_{\tilde{A}} = \hat{\sigma}_A / \hat{R}, \quad p_{\text{affine}} = \frac{\hat{A} - \hat{\delta} - \hat{R}A}{\hat{\sigma}_A}.$$

These treat the calibration table as frozen. Its offset/response covariance is retained separately for approximate propagated uncertainty. The calibration null scale $k_0 = \text{SD}[(\hat{A}(0) - \hat{\delta})/\hat{\sigma}_A]$ is also saved; dividing by it is a separate diagnostic, not part of the primary pull.



Mean raw pulls (dotted, faint) and frozen affine-corrected pulls (solid). Blue: nominal; orange: stress. Bars are standard errors. Each source uses its own calibration.



Corresponding sample pull widths. Bars are bootstrap standard errors. Zero mean and unit width are reference values; successful fits are not selected for agreement with them.

Calibration transfer across generators. Nominal calibration is additionally applied to independent stress toys at full exposure. This is a misspecification check, distinct from each generator's own conditional closure.

At $z = 3$, transferred affine pull means span $[-14.25, 12.43]$ and widths span 0.86–1.13.

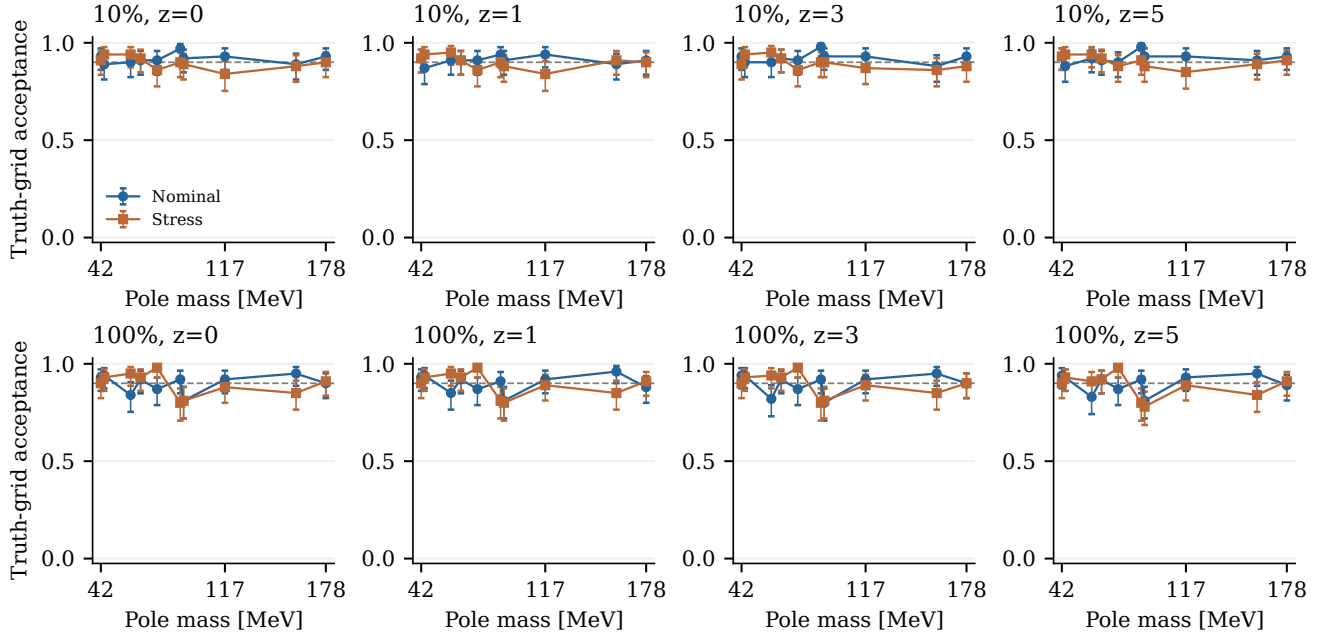
Centering one pinned generator is therefore insufficient evidence that the unknown physical background has been corrected.

A finite-rank Neyman construction

At each fixed grid value $A = z s_0$, $z = 0, \dots, 8$, use the raw signed fitted yield $T = \hat{A}$ as an ordering statistic. For 100 valid calibration toys, the lower-tail rank probability of a new spectrum is

$$p_A(T) = \frac{1 + \#\{j : T_{A,j}^{\text{cal}} \leq T\}}{101}.$$

Reject that grid value when $p_A \leq 0.10$. Exchangeability gives rejection probability at most 10/101 marginally over calibration and evaluation, with conservative ties. This is not a guarantee conditional on the one frozen calibration table.



Held-out acceptance of the injected grid value using the frozen same-generator table; exact 95% Clopper–Pearson intervals use each cell’s valid independent evaluation denominator. The dashed line is 90%. Intervals are pointwise; cells share backgrounds.

For continuous T , the population lower-tail mass below the tenth calibration order statistic has the Beta(10, 91) distribution. Its central 95% reference interval implies acceptance between 83.6% and 95.1% before observing the calibration table.

That order-statistic uncertainty is distinct from the held-out binomial interval and from the rank marginal size bound. It is not an extra confidence interval to add to a pull error.

At full-exposure 76 MeV, $z = 5$, same-source frozen acceptance is nominal 87/100 (95% CP [78.8%, 92.9%]) and stress 98/100 (95% CP [93.0%, 99.8%]). These cells need not realize exactly 90% acceptance.

The accepted set is retained as a discrete set. Its reported upper envelope is the largest accepted grid value; no interpolation or monotonic repair is applied. Empty sets, holes and acceptance of $z = 8$ (right censoring) are explicit. An empty set is assigned the physical reporting convention $U = 0$ and remains flagged; hence its upper bound contains $A = 0$ even if the rank test rejects that grid point. Acceptance of the true grid point differs from containment by this upper envelope. The result supplies no off-grid coverage guarantee.

Toy-CLs diagnostic and native profile limits

The finite-grid toy diagnostic uses the same raw-yield ordering and distributions from a calibration cohort independent of evaluation:

$$\text{CL}_s^{\text{toy}}(A) = \min\{1, p_A(T)/p_0(T)\}.$$

The same add-one rank probabilities are used in numerator and denominator. Reject at $\text{CL}_s^{\text{toy}} \leq 0.10$. Its rejection set is a subset of the rank- p_A rejection set and inherits the conservative matched-generator marginal bound. With 100 calibration toys, this ratio has finite Monte Carlo resolution and substantial tail uncertainty; it is a diagnostic, not a high-precision CLs calibration. Its accepted grid and censoring records are preserved without interpolation.

A finite-MC floor can hide failed transfer. At 76 MeV, full-exposure stress $z = 5$ toys with nominal calibration have 0/100 Neyman acceptance and 100 empty sets, yet toy CLs accepts 100/100 and has 100 right-censored endpoints. Here $p_A = p_0 = 1/101$ gives a ratio of one. This is finite-MC saturation, not successful calibration or a rescue of the native limit; native containment is also 0/100.

Method	Source	Exposure	Accept A	Envelope contains A	Median endpoint/ s_0
Neyman	Nominal	10%	88–98	88–98	4.00–4.00
Neyman	Nominal	100%	80–95	80–95	3.00–4.00
Neyman	Stress	10%	86–95	86–95	4.00–4.00
Neyman	Stress	100%	80–98	80–98	3.00–4.00
Toy CLs	Nominal	10%	88–98	88–98	4.00–4.00
Toy CLs	Nominal	100%	80–95	80–95	3.00–4.00
Toy CLs	Stress	10%	87–96	87–96	4.00–4.00
Toy CLs	Stress	100%	81–98	81–98	3.00–4.00
Native	Nominal	100%	77–100	77–100	3.77–4.99
Native	Stress	100%	0–100	0–100	0.20–16.77

At $z = 3$: ranges across ten masses, with counts out of each cell’s valid denominator (normally 100). Native acceptance and containment are the same upper-limit decision. A reported grid endpoint of 8 is a lower endpoint for a right-censored row, not a measured finite limit.

m [MeV]	Nominal $z = 3$	Nominal $z = 5$	Stress $z = 3$	Stress $z = 5$
42	89/100	86/100	0/100	0/100
44	94/100	94/100	0/100	0/100
60	93/100	91/100	1/100	1/100
66	91/100	89/100	100/100	100/100
76	77/100	73/100	0/100	0/100
90	100/100	100/100	30/100	28/100
92	93/100	91/100	9/100	8/100
117	100/100	100/100	97/100	95/100
160	95/100	92/100	99/100	97/100
178	88/100	89/100	73/100	69/100

Full-exposure native 90% profile-CLs upper-limit containment; each cell’s exact 95% Clopper–Pearson interval is supplied in `limit_summary.csv`.

The native calculation uses the original raw likelihood, with each toy’s GP mean/covariance fixed during nuisance profiling. Its denominator is the free fit for $\hat{A} \geq 0$ and the $A = 0$ fit otherwise; the GP mean is the null Asimov spectrum. The inherited bounded- $\hat{q}$ asymptotic tails give the root $\text{CL}_s = 0.10$. These full-exposure limits are never obtained by correcting a limit with $\hat{\delta}$ or $\hat{R}$. Toy-rank CLs and native profile CLs differ in ordering and approximation; agreement is an empirical result, not an identity.

Accounting, source boundaries and reproducibility

Set accounting. The following ranges are across all same-generator mass/strength cells. Counts record outcomes of 100 attempted evaluation toys per cell; they are not pooled rates. Complete per-cell valid counts, truth acceptance, upper-envelope containment, exact binomial intervals and set flags are retained.

Method	Source	Exposure	Valid	Empty	Holey	Right-censored
Neyman	Nominal	10%	100–100	0–11	0–0	0–15
Neyman	Nominal	100%	100–100	0–19	0–0	0–6
Neyman	Stress	10%	100–100	0–16	0–0	0–12
Neyman	Stress	100%	100–100	0–20	0–0	0–10
Toy CLs	Nominal	10%	100–100	0–0	0–0	0–15
Toy CLs	Nominal	100%	100–100	0–0	0–0	0–19
Toy CLs	Stress	10%	100–100	0–0	0–0	0–16
Toy CLs	Stress	100%	100–100	0–0	0–0	0–19

Generator provenance. The nominal generator is the pinned 2016 GP arithmetic mean. The archived conditional stress histogram retains the failed broad-component source-fit flag in its source manifest. Neither is an independently validated physical-background model. Background-source estimation uncertainty is not sampled. The exact 0.1 exposure branch here is not the historical 2016 subset: its archived support-count ratio is 0.1022, and exact luminosity fraction, selection equivalence and event overlap remain unverified.

Fitting and failures. The inherited signal resolution, pole-centered $\pm 2.25\sigma_m$ masks and reviewed mass-dependent kernel parameters remain fixed. Each spectrum recomputes count-dependent log-GP targets/noise, arithmetic mean and correlated covariance; no hyperparameter optimization is performed. The signed-yield Poisson likelihood profiles the same correlated GP nuisance constraint. Numerical retries retain the original spectrum and toy ID. Validity gates concern finite objectives, gradients and positive Poisson means, not closeness to truth or nominal coverage. Unresolved rows remain in the failure ledger.

The historical 2016 support/optimizer-qualification exception remains. Successful fixed-kernel conditional toys do not validate the original kernel optimization or repair that source qualification.

Frozen calibration uncertainty. Offset, response and their covariance come from calibration toys only. Whole-toy resampling preserves null/injected partners, masses and nested exposure pairs. The primary held-out pull treats the table as fixed; the separate approximate delta-method error also uses k_0 and is labeled accordingly; it does not recalibrate a likelihood interval. The finite-rank guarantee assumes exchangeable calibration/evaluation at a fixed generator; it does not establish physical-source coverage.

Reproduction. The package saves the numerical protocol, pilot reference, independent cohorts, calibration table, per-toy rows, summary tables, failed-row ledger, source hashes, vector figures and this LaTeX source. `scripts/analyze.py` computes summaries; `scripts/make_appendix.py -build` regenerates this appendix without fitting. The original 2021 PDF is copied and joined only during final packaging. `README.md` gives the validated commands. See the protocol and QA files for exact seeds, worker limits, completeness and final validation status.

The master seed is 63420160924; uncertainty summaries use 2,000 whole-toy bootstrap replicates. Calibration and evaluation bootstrap streams are separate. The deterministic mean-spectrum fits in `asimov_rows.csv` are not additional toys.

Claim boundary. This study can measure conditional offset scaling, response transfer and finite-toy performance under two pinned generators. It cannot establish unconditional coverage over unknown backgrounds, physical-model adequacy, calibrated discovery/global significance or an experimental exclusion.

Reference. Particle Data Group, *Statistics* (2026), sections 40.2 and 40.4.2, <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-statistics.pdf>. General bias and Neyman-construction context; the exposure approximation and finite-rank convention are defined explicitly above.